

Python Code

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1  from sklearn.datasets import fetch_20newsgroups
2  from sklearn.model_selection import train_test_split
3  from sklearn.feature_extraction.text import TfidfVectorizer
4  from sklearn.naive_bayes import MultinomialNB
5  from sklearn.metrics import accuracy_score, classification_report
6
7  categories = ['sci.space', 'rec.sport.hockey', 'comp.graphics', 'alt.atheism']
8
9  newsgroups = fetch_20newsgroups(subset='all', categories=categories, shuffle=True, r
andom_state=42)
10
11  print(f"Total documents: {len(newsgroups.data)}")
12  print(f"Target classes: {newsgroups.target_names}")
13
14  X_train, X_test, y_train, y_test = train_test_split(
15      newsgroups.data,
16      newsgroups.target,
17      test_size=0.2,
18      random_state=42
19  )
20
21  vectorizer = TfidfVectorizer(stop_words='english')
22  X_train_tfidf = vectorizer.fit_transform(X_train)
23  X_test_tfidf = vectorizer.transform(X_test)
24
25  nb = MultinomialNB()
26  nb.fit(X_train_tfidf, y_train)
27
28  y_pred = nb.predict(X_test_tfidf)
29
30  print("\n--- Model Evaluation ---")
31  print("Accuracy:", accuracy_score(y_test, y_pred))
32  print("\nClassification Report:\n")
33  print(classification_report(y_test, y_pred, target_names=newsgroups.target_names))
34
35  print("\n--- Sample Predictions ---")
36  for i in range(5):
37      print("\nText:\n", X_test[i])
38      print("Actual:", newsgroups.target_names[y_test[i]])
39      print("Predicted:", newsgroups.target_names[y_pred[i]])
```